Why OfficeBench tasks fail in Bangla

gemma4:31b (Ollama), temperature 0, 20-step limit · full run: 300 subtasks in each language, 297 compared pair by pair · 2026-10-03

Summary

The same model ran every OfficeBench subtask twice: once with the original English prompt and once with the system prompt, task and instructions translated into Bangla. Files, calendars and emails stayed in English in both runs. I compared the two runs subtask by subtask, read every subtask that passed in only one language, and classified the root cause of every Bangla failure.

- **Score.** Officially, English passes 132 of 300 (44.0%) and Bangla 103 of 300 (34.3%). With the email checks scored correctly (see the box below), on the 297 subtasks that can be compared, English passes **149 (50.2%)** and Bangla **127 (42.8%)**.
- **Most Bangla failures are not caused by Bangla.** Of the 170 Bangla failures, 130 also fail in English, and 89 of those fail for exactly the same reason: benchmark defects, running out of steps, cut-off output, wrong content.
- **40 failures are Bangla-only.** Over half of them (23) are the agent writing its result back in English the wrong way: answers in Bangla script (15), names in the wrong case (4), a different English word (3), a translated quoted phrase copied into a file (1). Another 4 come from the translation changing the meaning of the task. Only 6 are the agent reading or reasoning differently from a faithful translation, and 7 are execution problems.
- **Bangla also wins 18 subtasks** that English fails, mostly because English stepped into environment traps or wrote reasoning so long that its commands were cut off.
- **The gap closes with task difficulty:** single-app 63 vs 48, two-app 58 vs 51, three-app 28 vs 28.

About the scores. The repository's email check fails on the host machine for every run (apps/email_app/email_list_emails.py reads a hard-coded /testbed path), so the official result files undercount both languages. All numbers in this report use that check fixed and the mailbox matched by exact name; this also turns 3-12/0 into a pass in both languages. Excluded pairs: 2-1/0 and 2-1/1 (the model API returned empty replies in the Bangla run) and 3-52/0 (no English history).

The numbers

	Subtasks	English passes	Bangla passes	Gap
Single-app	93	63 (67.7%)	48 (51.6%)	15
Two-app	93	58 (62.4%)	51 (54.8%)	7
Three-app	111	28 (25.2%)	28 (25.2%)	0
All	**297**	**149 (50.2%)**	**127 (42.8%)**	**22**

Where the 170 Bangla failures come from

Group	Subtasks	What it means
Fail in both languages, same cause	89	Not a language effect: the agent or the benchmark fails the same way
Fail in both languages, different cause	41	Both fail, but Bangla fails differently (some of these are language effects, see Part 3)
Fail only in Bangla	40	Candidates for language effects (Part 1)
All Bangla failures	**170**	
For comparison: fail only in English	18	Subtasks Bangla passes and English fails (Part 2)

Part 1 — The 40 failures that happen only in Bangla

Cause	Subtasks	Language-related?
A. Result written back in English the wrong way	23	Yes
A1. Correct answer written in Bangla script	15	Yes
A2. Person’s name in the wrong case, or the wrong mailbox	4	Yes
A3. A different English word or format in the file	3	Yes
A4. A translated quoted phrase copied into the file	1	Yes (translation)
B. The translation changed what the task means	4	Yes (translation)
C. Faithful translation, but the task was read or reasoned about differently	6	Unclear
D. Execution: cut-off output, out of steps, "no tool"	7	Mostly no
Total	**40**	

A1. Correct answer written in Bangla script (15 subtasks)

The prompt never says which language to answer in, so the agent answers questions in Bangla, with Bangla digits. The checks look for English words or ASCII digits, so a correct answer is marked wrong. In all 15 cases the content matches the English run.

- 1-10/0 "what is the difference between the lowest and highest midterm-1 score?": "the highest is 100, the lowest is 60, the difference is 40", written in Bangla with Bangla digits.
- 1-13/2: answered "na" (Bangla for "no"); the check looks for "No".
- 2-31/0: answered "$20,000" with Bangla digits; the check looks for "20000".
- Others: 1-2/3, 1-2/4, 1-11/2, 1-12/0, 1-12/1, 1-23/1, 2-5/0, 2-32/0, 2-33/1, 2-34/1, 2-36/1, 2-36/2.
- Three more correct Bangla answers are hidden among the subtasks that fail in both languages (1-22/2, 1-22/3 "yes"; 1-22/4 "93"), where English got the answer wrong.

Why it happens: the agent writes to the user in the user’s language, but writes everything it passes to tools in English (no Bangla in the values written into cells, events and emails, apart from 3 runs: two copied the translated homework text (A4) and one wrote a congratulation letter to a student in Bangla). The benchmark only accepts English answers.

How to fix or test it: add "write the final answer in English" to the Bangla prompt, or score answers by meaning.

A2. Person’s name in the wrong case, or the wrong mailbox (4 subtasks)

In the email tasks about Bob, the Bangla task writes the name only in Bangla script, which has no capital letters. When the agent writes it back in Latin letters for the tool, it often writes "bob". Mailbox names are case-sensitive, so the listing is empty and the agent concludes there are no emails.

- 1-21/2 "does Bob receive emails in last month?": queried "bob", got "No emails found", answered "no" (in Bangla); the correct answer is "yes".
- 1-20/2 and 1-21/1: queried the user’s own mailbox ("Alice") instead of Bob’s and gave up.
- Others: 1-20/3.
- Measured over all runs: when a name appears in Latin letters in the prompt, the Bangla agent uses the exact case every time (49 of 49 tool calls; English 93 of 100). When the name appears only in Bangla script, it gets the case right in 26 of 37.

Why it happens: the agent has to rebuild an identifier from a script that has no case information.

How to fix or test it: keep person names in Latin script inside the Bangla task text; or make mailbox lookup case-insensitive.

A3. A different English word or format in the file (3 subtasks)

When the agent has to produce an English word itself, it sometimes picks a synonym or a different format from the one in the English task.

- 1-1/4 "Bob has a travelling ... set as a calendar event": event named "Trip to Florida"; the check looks for "travel".
- 1-4/1 "buy two yogurt": wrote "curd" (the Bangla word "doi" means both).
- 1-6/2 meetings "start from 8:00": wrote "08:00 AM"; the check looks for "8:00".
- Words the agent can copy from a file are never lost: "onion", "stock", "computer" and "dinner" all came back exactly, even when the Bangla prompt used a native word.

Why it happens: the Bangla prompt replaced the English word; the agent translates it back and may choose another word.

How to fix or test it: keep check keywords in Latin script inside the translation, or accept synonyms and equivalent formats.

A4. A translated quoted phrase copied into the file (1 subtask, plus 1 among shared failures)

The agent copies quoted text from the task exactly (17 of 17 quoted phrases in English were copied verbatim). Here the translator had translated the quoted text, so the agent faithfully copied the Bangla.

- 3-56/0 assign homework "read book and write essay": the Bangla task quotes it in Bangla, and the agent wrote the Bangla phrase into hw1.docx. The check looks for the English text.
- The same happens in 3-57/0, which also fails in English for another reason.

Why it happens: machine translation of a literal that should have stayed in English.

How to fix or test it: never translate quoted strings, file names or keywords.

B. The translation changed what the task means (4 subtasks)

In these cases the Bangla text says something different from the English, and the agent followed the Bangla correctly.

- 3-17/0 "make an image post welcome.jpg, email ... with welcome image in welcome.eml": the Bangla says "email the welcome image *that is contained in* welcome.eml". The agent searched for the files as if they already existed, then reported they could not be found.
- 3-55/0 "make a post reward.jpg": the Bangla says "make a post *for* reward.jpg". The agent wrote a social-media caption and never created the image. (English passed only by creating an empty file.)
- 3-68/0 "sum all the costs in costs.xlsx": the Bangla verb "jog korun" means both "add" and "sum". The agent added each cost as a row and never wrote the total.
- 2-40/2 "every event except sleeping": the Bangla says "except the act of sleeping". The agent also left out "nap" as a kind of sleep; the check needs "nap".

Why it happens: the task text was machine-translated by the same model, and several English tasks are terse or ungrammatical.

How to fix or test it: have the translations checked by a Bangla speaker before running.

C. Faithful translation, task read or reasoned about differently (6 subtasks)

The Bangla text means the same as the English, but the agent did something else. These are the closest thing to "reasoning in Bangla is worse", and they are few.

- 1-11/0 sort students by midterm1: wrote only the Name and midterm1 columns and dropped midterm2.
- 2-17/0 schedule meetings for each class: created the events in the user's own calendar instead of the students' calendars.
- 2-20/0 find students taking CS161: decided this meant "students whose Major is CS" and listed 3 of them.
- 2-38/2 send notification.eml to people: treated notification.eml as an existing file and switched between apps for 15 steps without sending anything (see the note on output files below).
- 2-32/1 compare income and expenditure: used the budget file as income and answered "larger"; English answered "smaller" for a wrong reason.
- 3-87/0 count qualifying terms: counted 11 at one step, then recounted at the next step and wrote 10.

Why it happens: no single cause. 3-87/0 is the known problem that earlier reasoning is not carried to the next step.

D. Execution problems (7 subtasks)

Problems that also happen in English, but happened here only in the Bangla run.

- **Cut-off output (4):** 1-15/0, 1-15/1, 2-6/0, 2-22/1. In 1-15/0 ("add a sentence to the end of the document") the Bangla agent assumed write_to_file overwrites the file and tried to rewrite the whole story; the response hit the 512-token limit and was retried 14 times. English wrote only the sentence (the tool appends) and passed.
- **Out of steps (2):** 3-64/0 and 3-9/0 spent 8 and 13 steps re-asking the in-container llm app, which answers in at most 50 tokens, for a complete text.
- **"No image tool" (1):** 3-15/0: "since I have no image-generation tool, I go straight to sending the email". English tried the shell's image converter and succeeded.

A pattern across groups B, C and the shared failures: output files read as inputs

In 7 Bangla runs the agent treated a file the task asks it to *create* as one that should already exist, and went looking for it: 2-38/2, 3-17/0, 2-44/0, 2-45/1, 3-12/2, 3-19/0, and 2-45/0 (which recovered and passed). The translations phrase these files with the Bangla definite marker, e.g. "notification.eml imel-ti pathan", meaning "send *the* email notification.eml", which presents the file as something that already exists. On the same tasks English either passed or failed for other reasons.

Part 2 — The 18 subtasks only Bangla passes

These explain why the gap is smaller than Part 1 suggests, and why it disappears on three-app tasks.

Why English failed and Bangla did not	Subtasks	Which
English stepped into an environment or check trap	7	2-34/0 (relative path), 2-36/3 and 3-89/0 (double quotes broke the shell command), 2-45/0 (one email to two addresses), 3-70/0 (.png instead of .jpg), 3-61/0 (unsupported attachment field), 2-39/0 (lower-case "bob"; Bangla retried "Bob")
English wrote long reasoning that pushed its command past the token limit, or wasted steps	7	2-43/0, 3-21/0, 3-59/0, 2-22/0 (cut-off), 3-26/0, 3-65/0, 2-8/0 (re-asking the llm app)
The translation settled an ambiguity the right way	2	1-8/0, 1-8/1 ("the header")
Bangla avoided an unnecessary action	1	1-2/1 (did not add a duplicate dinner event)
Bangla chose the expected word	1	3-92/0 (subject "Party Invitation")

Shorter reasoning in Bangla. On three-app tasks, Bangla reasoning averages 336 characters per step against 422 in English, and Bangla calls the 50-token llm app less often (151 vs 256 calls). English more often fills the 512-token response with reasoning before reaching the command (62 vs 19 responses cut off before the command). Runs with a malformed step: English 69, Bangla 28. This is why Bangla catches up on the hardest tasks.

No fake files in Bangla. On tasks that ask for an image, English created an empty placeholder image in 5 runs (e.g. 3-55/0 `touch reward.jpg`), which passes "file exists" checks. Bangla did this 0 times; when it could not make the image it said so. Both languages sometimes claim there is no image tool (14 English runs, 10 Bangla).

Part 3 — The 130 failures shared with English

These are the agent's and the benchmark's general problems. 89 fail for the same reason in both languages.

Root cause of the Bangla failure	1-app	2-app	3-app	Total	Same cause in English	English total
Benchmark defect decides the result	12	6	21	39	33	42
Wrong content (bad value, OCR misread, information dropped)	9	10	17	36	24	34
Out of steps (cell-by-cell writing, llm re-asking)	1	0	17	18	14	25
Task read differently	3	3	6	12	4	5
Cut-off output, retried	1	3	5	9	7	10
Empty result treated as "nothing exists"	1	4	4	9	5	9
Says the output is impossible ("no tool")	0	0	3	3	0	0
Finished with an output missing	0	2	0	2	2	2
Other	0	0	2	2	0	3
Total	**27**	**28**	**75**	**130**	**89**	**130**

3-12/0 is excluded from this table: it passes in both languages once the email check matches mailboxes by exact name.

Benchmark defects (39 Bangla failures)

A third of the shared failures cannot be passed by a correct agent, or depend on how ambiguous wording is read. Defects checked against the code:

- **Calendar times:** the calendar tool writes times without the "Z" suffix (DTSTART:20240531T130000), but several checks require it (3-20, 3-24, 3-28, 3-73, 3-74).
- **PDF names:** on Linux, convert_to_pdf calls LibreOffice with only an output folder, so the PDF keeps the source file's name while the tool reports the requested name (3-4, 3-67, 3-86, 3-25, 2-39/2).
- **Shell quoting:** every shell command is wrapped in `bash -c """..."""`, so a command containing double quotes silently writes nothing (2-33, 3-44, 3-72, 2-12).
- **Excel:** set_cell erases the value 1 (1-5/1, 1-6/0).
- **Wrong check targets or values:** 1-6/3 (wrong file), 1-14/2 (misspelled salery.xlsx), 1-14/3 (wrong reference), 3-1 (misspelled "Activaty"), 3-58 ("banana" not in the data), 3-8/5 (expects 7 and 4; the data gives 2 and 2), 2-2/0 and 2-2/1 (checks swapped), 3-62 (a PDF comparison that can never pass), 3-8/6 (the sender keeps a copy, so "Alice must not receive it" fails), 3-82 (docx opened as xlsx), 3-80 (wrong image name), 2-26/3, 3-99.
- **Checks asking for things the task does not:** 1-2/0, 1-2/2 (calendar events for a "find a time" task), 1-8/2, 1-8/4, 2-14, 2-45/2, 2-39/3, 3-43, 3-13/1; debatable: 1-6/4, 1-10/4, 3-22, 3-30, 3-49, 3-76.

Agent problems that appear in both languages

The rest are the agent's own habits, the same in English and Bangla:

- **Wrong content (36):** OCR misreads copied without checking (2-10/0: "Mice", "ob", "avid" written as names), labels dropped (2-23/0 wrote 104.98 without "total due"), numbers reformatted ("$1,200,000" for 1200000).
- **Out of steps (18):** the Excel tool writes one cell per action, so copying a table takes more than 20 steps (1-11/1, 3-35 to 3-41); or the agent keeps re-asking the 50-token llm app for a "complete" answer.
- **Cut-off output (9):** a whole document or long prompt in one action exceeds 512 tokens (1-19/1 retried from step 5 to 19).
- **Empty result read as "nothing exists" (9):** 3-90/0 lists only Bob's calendar, gets "No events found" and stops; the data is in Alice.ics.
- **Task read differently (12 Bangla vs 5 English):** Bangla has more of these, mostly the "output file read as input" pattern (2-12/0, 2-44/0, 2-45/1, 3-11/0, 3-12/2, 3-19/0) and 3-42/0, where the Bangla run never listed the data folder and made up a stock table.
- **"No image tool" (3, Bangla only):** 3-18/0, 3-85/0, 3-94/0. English failed these for other reasons, sometimes after creating an empty placeholder image.

Language effects hidden among the shared failures

Some subtasks fail in both languages, but the Bangla failure is still a language effect:

- 1-22/2, 1-22/3, 1-22/4: Bangla answered correctly ("yes", "yes", "93") in Bangla script; English answered wrongly after querying the mailbox "alice" in lower case.
- 3-57/0: the translated homework phrase was copied into the file (as in 3-56/0).
- 1-4/3: Bangla wrote "utility fee" (back from the Bangla "iutiliti fi"); English wrote "utilities"; the check wants "utilities fee".

Part 4 — What this means

- **Bangla does not break the agent's reasoning much.** Only 6 of 297 subtasks fail in Bangla because the agent read or reasoned differently from a faithful translation, and Bangla wins 18 subtasks back.
- **The cost of Bangla is at the boundaries.** 23 Bangla-only failures happen when the result has to be written back in English (answer language, name case, word choice, a translated literal), and 4 come from the translation itself. Over half of the gap would close with an English-answer instruction and better translations.
- **Most failures are shared.** Benchmark defects and the agent's general habits (one cell per action, long outputs cut off, no checking of its own work, giving up on empty results) cause most failures in both languages.
- **Bangla reasoning is shorter.** That costs nothing on easy tasks and helps on the hardest ones, where long English reasoning pushes commands past the 512-token limit.

Experiments that would confirm these causes

- Add "write the final answer in English" to the Bangla prompt (tests A1).
- Keep names, file names and check keywords in Latin script inside the Bangla tasks (tests A2, A3, A4 and the output-file pattern).
- Have the 300 translations checked by a Bangla speaker (tests B).
- Re-run the 58 subtasks that pass in only one language once more to confirm they are stable.
- Fix the benchmark defects in Part 3 and re-score both runs.

Appendix — every subtask that passes in only one language

ID	Task (short)	What went wrong in Bangla	Cause
1-2/3	Who gets up earlier?	Answered "Tom gets up earlier" in Bangla	A1
1-2/4	Who has more commitments?	Same answer as English, in Bangla	A1
1-10/0	Difference of lowest and highest score	"40" etc. in Bangla	A1
1-11/2	Monthly income	"35,000" in Bangla digits	A1
1-12/0	Students in CS161, how many seniors	"9 ... 2" in Bangla	A1
1-12/1	Students with 60 or more	"5" in Bangla	A1
1-13/2	Graduation requirement met?	"na" for "No"	A1
1-23/1	Freshmen in CS161	"4" in Bangla	A1
2-5/0	Application satisfies policy?	"yes" in Bangla	A1
2-31/0	Maximum car price in March	"$20,000" in Bangla digits	A1
2-32/0	Reimbursement over budget?	"yes" in Bangla	A1
2-33/1	Class with fewest students	"class 4" in Bangla	A1
2-34/1	Age with highest score	"26" in Bangla	A1
2-36/1	Dean’s list terms	"11" in Bangla	A1
2-36/2	Maximum sales entry	"78,002" in Bangla	A1
1-20/2	Latest email to Bob	Queried "Alice", gave up	A2
1-20/3	Did Bob receive emails?	Queried "bob", empty, "no, 0"	A2
1-21/1	Earliest email to Bob	Queried "Alice" then "bob"	A2
1-21/2	Did Bob receive emails?	Queried "bob", answered "no"	A2
1-1/4	Travel event	"Trip to Florida" (needs "travel")	A3
1-4/1	Buy two yogurt	"curd"	A3
1-6/2	Meetings from 8:00	"08:00 AM"	A3
3-56/0	Assign homework "read book..."	Homework text written in Bangla	A4
2-40/2	Email events except sleeping	Also left out "nap"	B
3-17/0	Image post welcome.jpg and email	Looked for the files as existing; none made	B
3-55/0	Statement and post reward.jpg	Wrote a caption "for" the image	B
3-68/0	Sum costs into costs.xlsx	Added rows, no total	B
1-11/0	Sort scores by midterm1	Dropped the midterm2 column	C
2-17/0	Meetings for each class	Events in the user’s calendar	C
2-20/0	Students taking CS161	Listed CS majors only	C
2-38/2	Send notification.eml	Looked for the .eml; sent nothing	C
2-32/1	Income vs expenditure	Used the budget as income	C
3-87/0	Dean’s list terms, email	Recounted 11 as 10	C
1-15/0	Add a sentence to a document	Rewrote the whole document; cut off	D
1-15/1	Append a section to the syllabus	Cut-off writes retried	D
2-6/0	Scan homework to docx	Cut-off writes retried	D
2-22/1	Instructions from price index	Cut-off writes retried	D
3-64/0	House price report with image	Out of steps (llm re-asking)	D
3-9/0	Revenue report as image	Out of steps (llm re-asking)	D
3-15/0	Contact card image from resume	"No image tool"	D

Subtasks only Bangla passes

ID	Bangla passes, English fails because ...
1-2/1	English added a duplicate dinner event that clashes with the existing one
1-8/0, 1-8/1	The Bangla wording of "the header" matched the reference
2-8/0	English re-asked the llm app 8 times and ran out of steps
2-22/0, 2-43/0, 3-21/0, 3-59/0	English reasoning re-listed long data and its command was cut off repeatedly
3-26/0, 3-65/0	English spent its steps re-asking the llm app
2-34/0	English wrote the file with a relative path, outside /testbed/data
2-36/3, 3-89/0	English used double quotes in a shell command, which the harness breaks
2-39/0	Both queried "bob"; Bangla retried "Bob" and found the emails
2-45/0	English sent one email to two addresses; the tool stored it for Bob only
3-61/0	English put the file name in an unsupported attachment field
3-70/0	English saved a .png; the check wants .jpg (the task does not say)
3-92/0	Bangla used the subject "Party Invitation"; English did not use the word "invitation"